

THE COMPLEXITY OF COUNTING CUTS AND OF COMPUTING THE PROBABILITY THAT A GRAPH IS CONNECTED*

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Abstract. Several enumeration and reliability problems are shown to be #P-complete, and hence, at least as hard as NP-complete problems. Included are important problems in network reliability analysis, namely, computing the probability that a graph is connected and counting the number of minimum cardinality (s, t) -cuts or directed network cuts. Also shown to be #P-complete are counting vertex covers in a bipartite graph, counting antichains in a partial order, and approximating the probability that a graph is connected and the probability that a pair of vertices is connected.

Key words. complexity, #P-complete, graphs, reliability, network reliability

1. Introduction. The inherent intractability of certain counting and reliability problems has been studied by Ball [1], Rosenthal [11], and Valiant [12]. Valiant defines the notion of the #P-complete class of counting problems, and shows that problems in this class are at least as hard as NP-complete problems. He then goes on to show that several important counting and reliability problems are #P-complete, among them, counting perfect matchings in bipartite graphs and evaluating the probability that two given nodes in a probabilistic graph are connected. Three important problems are mentioned by Ball and Valiant, for which the complexity is not known, namely:

- (1) evaluating the probability that a probabilistic graph is connected,
- (2) approximating the probability that a probabilistic graph is connected,
- (3) approximating the probability that two vertices of a probabilistic graph are connected.

In view of results by the authors in [2], the probability measure associated with problems (1) and (2) seems to have considerably more structure than that associated with (3). In [3] they also show the power of the structure in providing good upper and lower bounds for this measure. We show in this paper, however, that all three of these problems are NP-hard, in particular, #P-complete. In the process, we show that several counting problems are also #P-complete, among them: counting the number of node covers in a bipartite graph, counting antichains in a partial order, and counting minimum cardinality directed network cuts.

We now fix some terminology. Let $G = (V, E)$ be a graph with vertex set V and edge set E and let $m = |V|$ and $n = |E|$. When specified, G directed implies that the edges are taken to be ordered pairs, and G undirected implies the pairs are unordered. When not specified, G is allowed to be either. We allow loops (edges whose two end points are the same) and multiple edges (edges with the same pair of end points), although these are not strictly required for the results of this paper. Let s and t be two vertices in the graph G (directed or undirected). An (s, t) -path in G is any sequence $s = v_0, e_1, v_1, \dots, v_{k-1}, e_k, v_k = t$ of vertices $v_0, v_1, \dots$ and edges $e_1, e_2, \dots$ with $e_j = (v_{j-1}, v_j)$ for $j = 1, \dots, k$. An (s, t) -cut in G is any minimal set of edges that intersects every (s, t) -path. A network cut (with respect to s) is any minimal set of edges that is an (s, t) -cut for some vertex t in G . A spanning tree (rooted at s) is a

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minimal set of edges that contains paths from s to all other vertices in G . Note that if G is undirected, a network cut comprises any minimal set of edges whose removal disconnects G and a spanning tree is any minimal set of edges that connects all vertices; in both cases the definition is independent of the choice of s .

We now define our reliability measures. Given any real p , $0 \leq p \leq 1$, we impose the stochastic structure on G in which the edges of G are subject to random failure, independently and each with equal probability p . Edges that have not failed are said to be operative. We are concerned with two composite reliability measures on this stochastic model, which we will denote as functions of p . The first is the (s, t) *connectedness measure*: given vertices s and t in G ,

$$\begin{aligned} f(G, s, t; p) &= \Pr \{ \text{there is a path of operative edges from } s \text{ to } t \} \\ &= \Pr \{ \text{the failed edges of } G \text{ do not contain an } (s, t)\text{-cut} \}. \end{aligned}$$

The second is the *connectedness measure*: given vertex s in G ,

$$\begin{aligned} g(G, s; p) &= \Pr \{ \text{there is a path of operative edges from } s \text{ to every other vertex in } G \} \\ &= \Pr \{ \text{the failed edges of } G \text{ do not contain a network cut} \}. \end{aligned}$$

These measures are defined for both directed and undirected graphs. If G is undirected, then $f(G, s; p)$ is the probability that the operative edges in G form a connected graph on V , and is independent of the vertex s . The combinatorial significance of these reliability measures can be seen by expanding f and g :

$$\begin{aligned} f(G, s, t; p) &= \sum_{j=0}^n f_j p^j (1-p)^{n-j}, \\ g(G, s; p) &= \sum_{j=0}^n g_j p^j (1-p)^{n-j}, \end{aligned}$$

where

- f_j = number of sets of edges of cardinality j whose complement admits a path from s to t ,
- = number of sets of edges of cardinality j that do not contain an (s, t) -cut;
- g_j = number of sets of edges of cardinality j whose complement admits a path from s to every vertex in G ,
- = number of sets of edges of cardinality j that do not contain a network cut with respect to s .

The use of this form of the polynomial might seem slightly unnatural since coefficients are defined in terms of complements. However, it is consistent with the independence system interpretation of the reliability analysis problem used in other papers [2], [3]. Thus, the evaluation of f and g depend on the counting problems associated with (s, t) -cuts and network cuts in a way that will be shown precisely below.

We explore the computational complexity of counting and reliability problems in the manner proposed by Valiant [12]. The study of the complexity of feasibility and optimization problems has been pursued in the setting of recognition problems [5]. An important class is NP, which consists of those recognition problems accepted by a nondeterministic Turing machine of polynomial time complexity. The "hardest" problems in NP are called NP-complete; it is generally considered unlikely that polynomial algorithms exist for solving problems in this class. Valiant defines $\#P$ to be the set of integer-valued functions that can be computed by counting the number

of accepting computations of some nondeterministic Turing machine of polynomial time complexity. We extend Valiant's definition slightly to include rational and multiple valued functions that can be evaluated using functions of the above type. We say that a function f is *polynomially reducible* to a function g ($f \propto g$) if there exists an algorithm which, for any input z , evaluates $f(z)$ with a number of elementary operations and evaluations of g that is polynomial in the length of z . A function f is called $\#P$ -complete if (a) f is in $\#P$ and (b) every function g in $\#P$ can be reduced to f by a polynomial time reduction. The classes $\#P$ and $\#P$ -complete provide a natural setting for studying the complexity of counting and reliability problems. We remark that the counting problem associated with a given recognition problem is at least as hard as the recognition problem. In particular, the counting versions of NP-complete problems are NP-hard, i.e. at least as hard as NP-complete problems. To illustrate this point, note that a polynomial algorithm to determine the *number* of Hamiltonian circuits in a graph would immediately give a polynomial algorithm to determine if a graph contained at least one Hamiltonian circuit. In fact, the counting versions of most NP-complete problems can be easily shown to be $\#P$ -complete. See [5] for a detailed treatment of NP-completeness and its relationship to $\#P$ -completeness.

With these definitions in mind we state our main result:

THEOREM. *The following functions are $\#P$ -complete:*

1. **BIPARTITE VERTEX COVER**
Input: bipartite graph $G = (V, E)$
Output: $|\{S \subseteq V: \text{for each } e = (u, w) \in E, u \in S \text{ or } w \in S\}|$;
2. **BIPARTITE INDEPENDENT SET**
Input: bipartite graph $G = (V, E)$
Output: $|\{S \subseteq V: \text{for all } u, w \in S, e = (u, w) \notin E\}|$;
3. **ANTICHAIN**
Input: partial order $(X, \leq)$
Output: $|\{S \subseteq X: \text{there are no } x, y \in S \text{ with } x \leq y\}|$;
4. **MINIMUM CARDINALITY BIPARTITE VERTEX COVER**
(MAXIMUM CARDINALITY BIPARTITE INDEPENDENT SET,
MAXIMUM CARDINALITY ANTICHAIN, RESPECTIVELY)
Input: same as 1 (2, 3, resp.)
Output: the number of minimum cardinality (maximum cardinality, resp.) elements of the output set;
5. **BIPARTITE 2-SAT WITH NO NEGATIONS**
Input: Boolean expression B in the variables $x_1, \dots, x_k, y_1, \dots, y_l$ of the form $B = (x_{i_1} \vee y_{j_1}) \wedge \dots \wedge (x_{i_n} \vee y_{j_n})$
Output: $|\{x_1, \dots, x_k, y_1, \dots, y_l \text{ that satisfy } B\}|$;
6. **MINIMUM CARDINALITY (s, t) -CUT**
Input: graph $G = (V, E), s, t \in V$
Output: $|\{C \subseteq E: C \text{ is a minimum cardinality } (s, t)\text{-cut in } G\}|$;
7. **MINIMUM CARDINALITY DIRECTED NETWORK CUT**
Input: directed graph $G = (V, E), s \in V$
Output: $|\{C \subseteq E: C \text{ is a minimum cardinality network cut with respect to } s\}|$;
8. **CONNECTEDNESS RELIABILITY**
Input: graph $G = (V, E), s \in V$, rational $p, 0 \leq p \leq 1$
Output: $g(G, s; p)$;
9. **CONNECTEDNESS RELIABILITY ϵ -APPROXIMATION**
Input: graph $G = (V, E), s \in V, \epsilon \geq 0$, rational $p, 0 \leq p \leq 1$
Output: rational r with $r - \epsilon < g(G, s; p) < r + \epsilon$;

10. (s, t) CONNECTEDNESS RELIABILITY ε -APPROXIMATION

Input: graph $G = (V, E)$, $s, t \in V$, $\varepsilon > 0$, rational p , $0 \leq p \leq 1$

Output: rational r with $r - \varepsilon < f(G, s, t; p) < r + \varepsilon$;

Before going on to the proof of the theorem, we illustrate how our results fit in with previous results concerning reliability and important related counting problems. Computation of the functions f and g are considered the two most important and well-studied network reliability problems. The theorem settles the complexity of computing g exactly and the ε -approximation problem for f and g . In terms of computing or approximating f , two important quantities are the number of minimum cardinality (s, t) -cuts and the number of minimum cardinality (s, t) -paths. These correspond, respectively, to the first $f_i < \binom{n}{i}$ and the last $f_i > 0$. The two corresponding quantities for g are the number of minimum cardinality network cuts and the number of minimum cardinality connected sets, i.e. spanning trees, and these correspond, respectively, to the first $g_i < \binom{n}{i}$ and the last $g_i > 0$. Table 1 describes the known

TABLE 1

	Min. card. pathset	Min. card. cutset	Rel. poly.	Rel. approx.
undirected and directed two-terminal (f)	* [3]	! TH	! [12]	! TH
undirected network (g)	* [10] [†]	* [3]	! TH [§]	! TH
directed network (g)	* [10] [†]	! TH	! TH [¶]	! TH

Either the appropriate reference is given or TH which indicates the result is contained in the theorems given in this paper; * implies polynomial; ! implies #P-complete.

[†] Reference [10] reduces the problem to computing the determinant of a matrix. It is (now) well known that determinants can be computed in polynomial time.

[§] These results have recently been proven independently by Jerrum [9].

[¶] This result has recently been proven independently by Hagstrom [6].

complexity results for all of these problems. It uses the generic term pathsets to refer to both spanning trees and (s, t) -paths and cutsets refer to both (s, t) -cuts and network cuts. Columns 1 and 2 refer to the problems of determining the number of minimum cardinality pathsets and cutsets respectively, column 3 to the problem of determining the polynomial f or g , and column 4 to the approximation problem defined in parts 9 and 10 of the theorem.

2. Proof of the theorem. The format for establishing a function f as #P-complete is as follows. We first establish that f is in #P by showing that, for any input z , there exists a polynomial algorithm for recognizing structures associated with the input z whose number is $f(z)$. In the context of the functions given in the theorem this is a trivial matter, since virtually all the functions count easily recognizable objects in the graph G associated with the input z . To show that f is #P-complete, we start with a known #P-complete function g , and show that there exists an algorithm which, for any z , evaluates $g(z)$ using a polynomial number of evaluations of f . In many cases this simply involves altering the input z (here the graph G) in polynomial time to a new input z' (here a new graph G') for which $g(z) = f(z')$. In some cases, however, we must evaluate f for a number of inputs $z_1, \dots, z_n$, that number being polynomial in the size of z . We then relate the values $f(z_i)$, $i = 1, \dots, n$ to the value of g by

equations of the form

$$(1) \quad v_i = f(z_i) = \sum_{j=1}^k a_{ij} b_j, \quad i = 1, \dots, k$$

where the a_{ij} are known and $g(z)$ is some simple function of the b_j . If we can show that the $k \times k$ matrix of the coefficients a_{ij} for (1) is nonsingular, we can perform k evaluations of f , and then solve the linear system to obtain the values of b_j , and hence the value of $g(z)$.

Valiant, in [12], has made use of a special class of matrices to produce the desired nonsingular systems discussed above. A *Vandermonde matrix* is an $(n+1) \times (n+1)$ matrix of the form

$$\Delta = \begin{pmatrix} 1 & \mu_0 & \mu_0^2 & \cdots & \mu_0^n \\ 1 & \mu_1 & \mu_1^2 & \cdots & \mu_1^n \\ \vdots & & & & \\ 1 & \mu_n & \mu_n^2 & \cdots & \mu_n^n \end{pmatrix}$$

(or its transpose), where $\mu_0, \dots, \mu_n$ are arbitrary real numbers. A well-known fact about these matrices (see, for example [7, § 5.1]) is $\det \Delta = \prod_{i>j} (\mu_i - \mu_j)$. We have immediately from the previous discussion:

LEMMA. Suppose we have v_i and b_i , $i = 1, \dots, n+1$, related by the equation

$$v_i = \sum_{j=1}^{n+1} a_{ij} b_j, \quad i = 1, \dots, n+1.$$

Further, suppose that the matrix of coefficient (a_{ij}) is Vandermonde, with parameters $\mu_0, \dots, \mu_n$ which are distinct. Then, given values for $v_1, \dots, v_{n+1}$, we can obtain the values $b_1, \dots, b_{n+1}$ in time polynomial in n .

We will make repeated use of this lemma throughout the proof of the theorem.

It is easy to see that the problems 1–10 of the theorem are in #P. To show that they are #P-complete, we establish a sequence of reductions, starting with the following counting problem.

CARDINALITY VERTEX COVER

Input: graph $G = (V, E)$, integer k

Output: $|\{S \subseteq V : S \text{ is a vertex cover for } G \text{ and } |S| = k\}|$.

This problem is known to be #P-complete (see [5, p. 169]). We also define one intermediate problem for purposes of the proof, namely,

0. VERTEX COVER

Input: graph $G = (V, E)$

Output: $|\{S \subseteq V : \text{for each } e = (u, v) \in E, u \in S \text{ or } v \in S\}|$.

We now give the reductions.

0. CARDINALITY VERTEX COVER $\propto$ VERTEX COVER. Given $G = (V, E)$, for $l = 1, \dots, m = |V|$, construct graph $G'(l)$ with vertex set $V'(l) = \{v'_i : v \in V, i = 1, \dots, l\}$ and edge set $E'(l) = \{(u'_i, v'_j) : (u, v) \in E, i = 1, \dots, l, j = 1, \dots, l\}$. This construction is illustrated in Fig. 1. Now every cover C' of $G'(l)$ has the property that if $(u, v) \in E$ then $\{u'_1, \dots, u'_l\} \subseteq C'$ or $\{v'_1, \dots, v'_l\} \subseteq C'$. Therefore, for each cover C of G there corresponds a class $\Omega(C)$ of covers of $G'(l)$ with elements of the form $\bigcup_{v \in V} S'_v$, where $S'_v = \{v'_1, \dots, v'_l\}$ if $v \in C$ and $S'_v \subseteq \{v'_1, \dots, v'_l\}$ if $v \notin C$. The class $\Omega(C)$ consists of $(2^l - 1)^{m-|C|}$ covers, and the classes $\{\Omega(C) : C \text{ a cover of } G\}$ partition

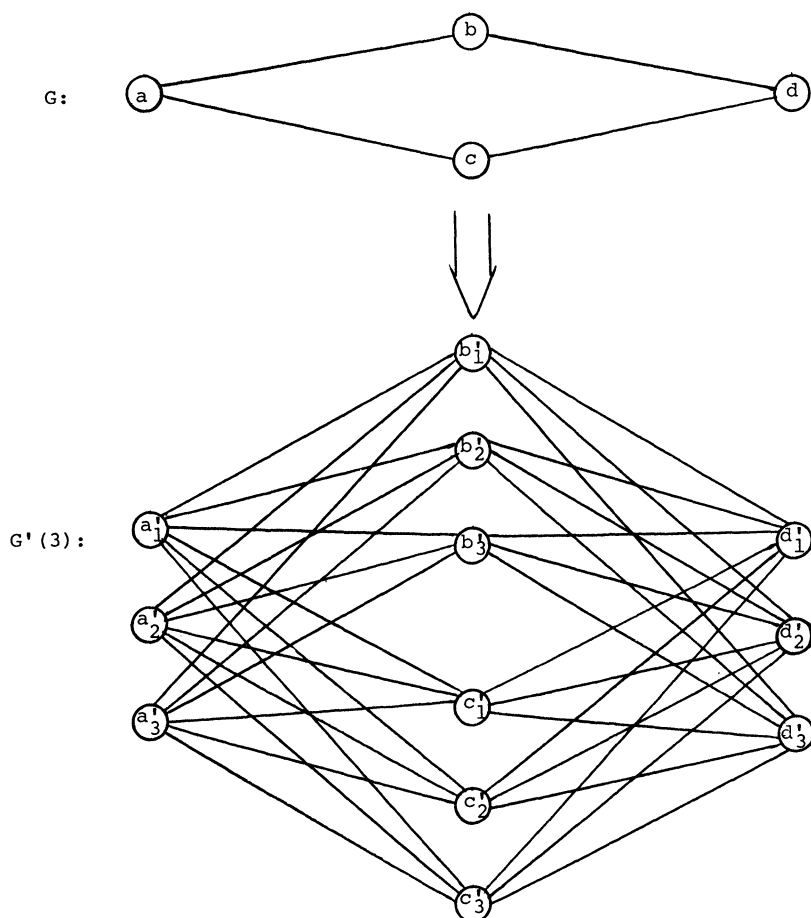


FIG. 1. Example of transformation used in reduction 0.

the covers of $G'(l)$. The number of covers of $G'(l)$ is therefore

$$(2) \quad \Gamma(l) = \sum_{i=0}^{m-1} A_i (2^l - 1)^i,$$

where A_i is the number of covers of G of cardinality $m - i$, $i = 0, \dots, m - 1$. Now the $m \times m$ matrix $B = (b_{jl})$ with entries $b_{jl} = (2^l - 1)^{j-1}$ $j = 1, \dots, m$, $l = 1, \dots, m$, is Vandermonde with $\mu_l = 2^l - 1$ distinct for $l = 1, \dots, m$. Therefore, by the lemma we can solve (2) to obtain each A_i , and hence solve the cardinality vertex cover problem.

1. VERTEX COVER $\propto$ BIPARTITE VERTEX COVER. Given $G = (V, E)$, for $l = 0, \dots, N = \binom{m+2}{2} - 1$ construct bipartite graph $G'(l)$ by replacing each edge (u, v) in G by the subgraph shown in Fig. 2. (Note that when $l = 0$, the graph $G'(l)$ has no edges at all.) This subgraph has the property that the number of vertex covers containing neither u nor v is 2^l , the number of covers containing a particular one of u or v is 3^l and the number of covers containing both u and v is 5^l . Thus, the number of covers of $G'(l)$ is

$$(3) \quad \Gamma(l) = \sum_{\substack{i+j+k=n \\ i,j,k \geq 0}} A_{ijk} (2^l)^i (3^l)^j (5^l)^k = \sum_{\substack{i+j+k=n \\ i,j,k \geq 0}} A_{ijk} (2^i 3^j 5^k)^l,$$

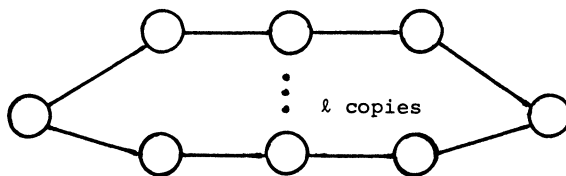


FIG. 2

where A_{ijk} is the number of sets S of vertices in G for which i edges of G have neither vertex in S , j edges have exactly one vertex in S , and k edges have both vertices in S . The $N \times N$ matrix $B = (b_{ql})$ defined

$$b_{ql} = (2^{i_q} 3^{j_q} 5^{k_q})^l, \quad q = 1, \dots, N, \quad l = 0, \dots, N-1,$$

where (i_q, j_q, k_q) are all triples summing to n , is Vandermonde. Further, $\mu_q = 2^{i_q} 3^{j_q} 5^{k_q} = 2^{i_r} 3^{j_r} 5^{k_r} = \mu_r$ if and only if $i_q = i_r$, $j_q = j_r$, and $k_q = k_r$. Therefore, the μ_q are distinct and by the lemma we can solve (3) to obtain each A_{ijk} , for $i + j + k = n$, $i \geq 0$, $j \geq 0$, $k \geq 0$. In particular, we can obtain

$$A = \sum_{\substack{j+k=n \\ j,k \geq 0}} A_{0jk},$$

which is the number of sets of vertices of G for which no edge of G is uncovered, that is, the number of covers of G .

2. BIPARTITE VERTEX COVER $\propto$ BIPARTITE INDEPENDENT SET. Given $G = (V, E)$ we note that $C \subseteq V$ is a cover for G if and only if $V - C$ is an independent set in G . The reduction follows.

3. BIPARTITE INDEPENDENT SET $\propto$ ANTICHAIN. Given bipartite graph $G = (V, E)$ with $V = V_1 \cup V_2$ and $E \subseteq V_1 X V_2$, define partial order $(X, \leq)$ with $X = V$ and order defined for $x \neq y \in X$: $x \leq y$ if and only if $x \in V_1$, $y \in V_2$, and $(x, y) \in E$. $(X, \leq)$ is trivially transitive and antisymmetric. Further, a set $S \subseteq X$ is an antichain in $(X, \leq)$ if and only if it is independent in G . The reduction follows.

4. BIPARTITE VERTEX COVER $\propto$ MINIMUM CARDINALITY BIPARTITE VERTEX COVER (MAXIMUM CARDINALITY BIPARTITE INDEPENDENT SET, MAXIMUM CARDINALITY ANTICHAIN, RESPECTIVELY). Given bipartite graph $G = (V, E)$, construct bipartite graph $G' = (V', E')$ by adding vertices $\{v' : v \in V\}$ to V and pendant edges $M = \{(v, v') : v \in V\}$ to E . Now since M consists of m disjoint edges that cover all vertices of G (a perfect matching), it follows that a minimum cardinality vertex cover of G' is of cardinality m . Furthermore, there is a 1-1 correspondence between vertex covers of G and minimum cardinality vertex covers of G' obtained by associating with cover C of G the cardinality m cover

$$C' = \{v : v \in C\} \cup \{v' : v \notin C\}.$$

In view of the discussion in reductions 2 and 3, it follows easily that the bipartite vertex cover problem reduces to any of the three given minimum or maximum cardinality problems.

5. BIPARTITE VERTEX COVER $\propto$ BIPARTITE 2-SAT WITH NO NEGATIONS. Given bipartite graph $G = (V, E)$ with $V = V_1 \cup V_2$, $V_1 = \{u_1, \dots, u_k\}$,

$V_2 = \{v_1, \dots, v_l\}$ define Boolean expression in $x_1, \dots, x_k, y_1, \dots, y_l$ by

$$f(x_1, \dots, x_k, y_1, \dots, y_l) = \bigwedge_{e=(u_i, y_j) \in E} (x_i \vee y_j).$$

Then $f(x_1, \dots, x_k, y_1, \dots, y_l)$ is true if and only if $\{u_i : x_i = T\} \cup \{v_j : y_j = T\}$ forms a cover of G . The reduction follows.

6. BIPARTITE INDEPENDENT SET $\propto$ MINIMUM CARDINALITY (s, t) -CUT. Given a bipartite graph $G = (V, E)$, $V = V_1 \cup V_2$, $E \subseteq V_1 \times V_2$, construct the graph G' with vertices $V \cup \{s, t\}$ and edges consisting of E along with sets M'_v of multiple edges of the type (s, v) , $v \in V_1$ or (v, t) , $v \in V_2$ with multiplicity equal to the degree of v in G . An example of this construction is given in Fig. 3. Now a minimum

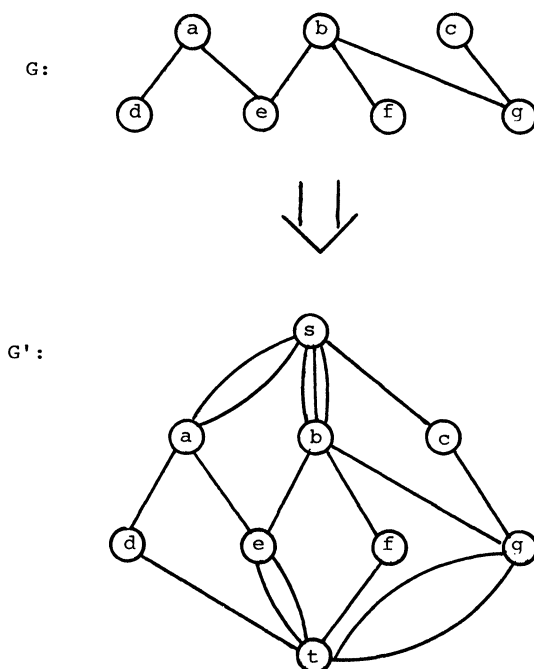


FIG. 3. Example of transformation used in reduction 6.

cardinality (s, t) -cut in G' is of cardinality $|E|$, since (a) E is an (s, t) -cut and (b) an (s, t) -flow of size $|E|$ can be obtained by directing all edges from s to t and giving each a flow of 1. It is clear that if a minimum cardinality (s, t) -cut C' of G' contains one edge of a set M'_v then it must contain every edge in M'_v . Further, if the edges (s, v) and (w, t) are in C' , then (v, w) cannot be an edge, since we can obtain a cut with one less edge by replacing M'_v and M'_w with all edges in E adjacent to either v or w . Thus, the sets M'_v of C' have ends in G which are independent in G and the remaining edges in C' must be all those edges in E which do not have a vertex in common with these sets. Conversely, any set of edges of this type must be a minimum cardinality (s, t) -cut in G' . Thus, there is a one to one correspondence between minimum cardinality (s, t) -cuts in G' and independent sets in G . The reduction is now complete. Note that this reduction applies in both the directed and undirected cases.

The use of multiple edges could have been avoided but we omit the argument for the sake of simplicity.

7. DIRECTED MINIMUM CARDINALITY (s, t) -CUT $\propto$ MINIMUM CARDINALITY DIRECTED NETWORK CUT. Given directed graph $G = (V, E)$, $s, t \in V$, let k be the cardinality of a minimum cardinality (s, t) -cut. (It is well known that k can be calculated in polynomial time using a network flow algorithm.) Construct directed graph G' from G by adding multiple edges of the form (t, v) with multiplicity $k + 1$ for each $v \in V - \{s, t\}$. Fig. 4 illustrates this transformation. Now any

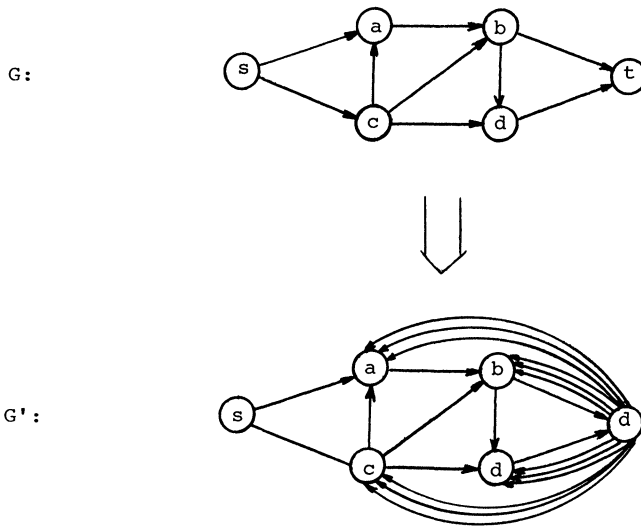


FIG. 4. Example of transformation used in reduction 7.

minimum cardinality (s, t) -cut in G remains a network cut in G' since all of the added edges point out of t . Thus, the size of a minimum cardinality network cut in G' is at most k . But since removal of any set S of at most k edges from E' must leave at least one edge from t to every vertex $x \neq s$ in V , then S is a network cut in G' if and only if S is an (s, t) -cut in G . Therefore, the minimum cardinality network cuts for G' are of cardinality k , and they consist precisely of sets of edges which are (s, t) -cuts for G . This completes the reduction. As in the previous argument, the use of multiple edges could have been avoided.

8A. MINIMUM CARDINALITY DIRECTED NETWORK $\propto$ DIRECTED CONNECTEDNESS RELIABILITY. Given $G = (V, E)$, we write, as in § 1,

$$(4) \quad g(G, s; p) = \sum_{j=0}^n g_j p^j (1-p)^{n-j} = (1-p)^n \sum_{j=0}^n g_j \left(\frac{p}{1-p} \right)^j,$$

where g_j is the number of sets of edges of cardinality j whose complement admits a path from s to every other vertex in G . Thus, $\bar{g}_j = \binom{n}{j} - g_j$ is the number of sets of edges of cardinality j that contain a directed network cut. Further, the matrix $B = (b_{ij})$ with $b_{ij} = (p_i/(1-p_i))^j$ for $i = 0, \dots, m$, $j = 0, \dots, n$ is Vandermonde for any choice $0 < p_0 < \dots < p_n < 1$. Therefore, by evaluating $g(G, s; p)/(1-p)^n$ for $i = 0, \dots, n$, and

solving (4) we can obtain g_j , and hence $\bar{g}_j$ for $j = 0, \dots, n$. The value of the first nonzero $\bar{g}_j$ then solves the minimum cardinality directed network cut problem.

8B. MINIMUM CARDINALITY UNIDIRECTED (s, t) -CUT $\propto$ UNIDIRECTED CONNECTEDNESS RELIABILITY. Given undirected graph G , vertices s, t , write the network reliability polynomial of G with respect to s as above

$$g(G, s; p) = (1-p)^n \sum_{i=0}^n g_i \left(\frac{p}{1-p} \right)^i,$$

where g_i is the number of sets of cardinality i whose complement admits a path from every vertex to s . Consider now the graph G' obtained from G by replacing the vertices s and t with the vertex v'_{st} in every edge in which either appears. The network reliability polynomial of G' with respect to v'_{st} is

$$g(G', v'_{st}; p) = \sum_{i=0}^n g'_i p^i (1-p)^{n-i} = (1-p)^n \sum_{i=0}^n g'_i \left(\frac{p}{1-p} \right)^{n-i}.$$

Now g'_i is the number of sets of edges in G' of cardinality i whose complement admits a path from every vertex of G' to v'_{st} , or equivalently, the number of sets of edges in G of cardinality i whose complement admits a path from every vertex of G to either s or t . Therefore, $g'_i - g_i$ is the number of sets of edges in G of cardinality i whose complement admits a path from every vertex to s or t but does not admit a path from every vertex to both s and t . Such a set in particular contains an (s, t) -cut. Let k be the cardinality of a minimum cardinality (s, t) -cut. Then the complement of any set of k edges that contains an (s, t) -cut must allow a path from every vertex to either s or t (otherwise, an edge could be added to the component containing a vertex not connected to either s or t and still not allow a path from s or t). Thus $g'_k - g_k$ is the number of minimum cardinality (s, t) -cuts in G . As in problem 8A, by evaluating $g(G, s; p_i)$ and $g(G', v'_{st}; p_i)$ for $0 < p_0 < \dots < p_n$, we can obtain g_i and g'_i for $i = 0, \dots, n$, and in particular, the value $g'_k - g_k$. This completes the reduction.

9. MINIMUM CARDINALITY NETWORK CUT $\propto$ CONNECTEDNESS RELIABILITY APPROXIMATION. Suppose we are given $G = (V, E)$ and $s \in V$. We produce this reduction by showing how to compute the g_i successively for $i = 0, 1, \dots$, using as a subroutine an algorithm for the connectedness reliability approximation problem. Suppose we have computed g_i for $i = 0, 1, \dots, k-1$; define

$$\alpha = \sum_{j=0}^{k-1} g_j p^j (1-p)^{n-j};$$

then for $0 < p < 1$ we have

$$\begin{aligned} g(G, s; p) - \alpha &= \sum_{j=k}^n g_j p^j (1-p)^{n-j} \\ &= p^k (1-p)^{n-k} \left[g_k + \frac{p}{1-p} \sum_{i=k+1}^n g_i \left(\frac{p}{1-p} \right)^{i-k-1} \right]. \end{aligned}$$

Using the fact that $0 \leq g_i \leq \binom{n}{i}$ for $i = k+1, \dots, n$, we obtain the inequalities

$$\frac{g(G, s; p) - \alpha}{p^k (1-p)^{n-k}} \geq g_k$$

and

$$\begin{aligned}
 & \frac{g(G, s; p) - \alpha}{p^k (1-p)^{n-k}} \\
 & \leq g_k + \frac{p}{1-p} \sum_{i=k+1}^n \binom{n}{i} \left(\frac{p}{1-p}\right)^{i-k-1} \\
 & = g_k + \frac{p}{1-p} \sum_{i=k+1}^n \left[\frac{n!}{(n-k-1)!} \right] \left[\frac{(i-k-1)!}{i!} \right] \left[\left(\frac{p}{1-p}\right)^{i-k-1} \right] \\
 & \leq g_k + \frac{p}{1-p} \left[\frac{n!}{(n-k-1)!} \right] \left[\frac{1}{(k+1)!} \right] \left[\frac{1}{(1-p)^{n-k-1}} \right] \\
 & = g_k + \binom{n}{k+1} \frac{p}{(1-p)^{n-k}}.
 \end{aligned}$$

Now if r is an ε -approximation to g , it follows for $0 < p < 1$ that

$$\begin{aligned}
 g_k & \leq \frac{r + \varepsilon - \alpha}{p^k (1-p)^{n-k}} = \frac{(r - \varepsilon) - \alpha + 2\varepsilon}{p^k (1-p)^{n-k}} \leq \frac{g(G, s; p) - \alpha + 2\varepsilon}{p^k (1-p)^{n-k}} \\
 & \leq g_k + \binom{n}{k+1} \frac{p}{(1-p)^{n-k}} + \frac{2\varepsilon}{p^k (1-p)^{n-k}} \\
 & = g_k + \frac{1}{(1-p)^{n-k}} \left[\binom{n}{k+1} p + \frac{2\varepsilon}{p^k} \right],
 \end{aligned}$$

so that, if we choose

$$p = \min \left\{ 1 - 2^{-1/(n-k)}, \frac{1}{2} \binom{n}{k+1}^{-1} \right\}$$

and $\varepsilon = p^k/4$, then

$$\frac{1}{(1-p)^{n-k}} \left[\binom{n}{k+1} p + \frac{2\varepsilon}{p^k} \right] < \frac{1}{1/2} \left[\binom{n}{k+1} \frac{1}{2} \binom{n}{k+1}^{-1} + \frac{p^{k/2}}{p^k} \right] = 2 \left[\frac{1}{2} + \frac{1}{2} \right] = 1.$$

Hence,

$$g_k = \left\lfloor \frac{r + \varepsilon - \alpha}{p^k (1-p)^k} \right\rfloor.$$

The proof is now complete.

10. MINIMUM CARDINALITY (s, t) -CUT $\propto (s, t)$ -CONNECTEDNESS RELIABILITY APPROXIMATION. The reduction here is identical to that in problem 9. This completes the proof of the theorem.

3. Further discussion. We remark that problems 9 and 10 easily show the #P-completeness of the α -approximation problem (see [11], called the *point estimate* problem in [1] for the functions g and f). This problem is: given $\alpha < 1$, $0 \leq p \leq 1$, find a number r such that $\alpha r < g(G, s; p)$ (respectively $f(G, s, t; p) < r/\alpha$). We should note that a seemingly more difficult unsolved problem involves the case where α (or ε) is constant, i.e. is not allowed to vary as part of the input list.

We complete our discussion by considering the complexity of certain reliability and counting problems for two special classes of graphs. One class is that of directed

acyclic graphs, that is, graphs that have no closed (directed) paths. For these graphs the minimum cardinality (s, t) -cut problem (6) still remains $\#P$ -complete since the network constructed in the proof of the theorem is acyclic; hence, the (s, t) -connectedness problem (10) for acyclic graphs remains $\#P$ -complete. The directed network cut problem (7), however, is polynomial, and, in fact, the connectedness reliability problems (8 and 9) are also polynomial (see [3]). The second class of graphs is that of planar graphs (directed and undirected). Here, both the minimum cardinality network cut problem and the minimum cardinality (s, t) -cut problem are polynomial (see also [3]). The complexity of the reliability problems, however, are open questions. Table 2 summarizes known results for these classes of graphs.

TABLE 2

	Min. card. pathset	Min. card cutset	Rel. poly.	Rel. approx.
directed acyclic two terminal	* [3]	! TH	! TH	! TH
directed acyclic network	* [3]	* [3]	* [3]	* [3]
undirected and directed planar two terminal	* [3]	* [3]	?	?
undirected and directed planar network	* [10]	* [3]	?	?

The table entries have the same interpretation as those in Table 1.

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